

EAP

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EAP: Architecture Specification and Prototype

Creating a space where developers can quickly find accurate, reliable answers to their coding questions. What sets it apart is the community-driven approach, where users contribute their expertise, and the platform's voting and reputation system rewards quality contributions. We hope to create a self-sustaining cycle of knowledge.

A7: Web Resources Specification

This artifact show not only the overview of features that the platform will have, but also different permissions and the openapi file.

1. Overview

Module	Description
M01: Authentication	Web resources associated with the authentication, including login, logout and register actions.
M02: Individual Profile and User Administration	Web resources associated with individual profile, including view and edit personal information and management of users, as view, block or delete users.
M03: Publications	Web resources associated with add, view, search, edit and delete questions and answers.
M04: Static pages	Web resources associated with static pages, such as about us, main features and contacts.
M05: Reports	Web resources associated with the reports, including reporting questions and answers.
M06: Notifications	Web resources associated with the notifications, like receive and view notifications.

2. Permissions

PUB	Public	Unregistered users and without privileges
USR	User	Authenticated users with privileges
OWN	Owner	Users that are owners of some information (e.g questions, answers,reports, profile)
ADM	Administrator	Platform administrators with special privileges (eg. ban other users)

3. OpenAPI Specification

[a7_openapi.yaml](#).

```
openapi: 3.0.0

info:
  version: '1.0'
  title: 'LBAW ASKIT Web API'
  description: 'Web Resources Specification (A7) for ASKIT'

tags:
- name: 'M01: Authentication'
- name: 'M02: Individual Profile and User administration'
- name: 'M03: Publications'
- name: 'M04: Static pages'
- name: 'M05: Reports'
- name: 'M06: Notifications'

paths:
  # MODULE 01: Authentication
  /login:
    get:
```

```
operationId: R101
summary: 'R101: Login Form'
description: 'Provide login form. Access: PUB'
tags:
  - 'M01: Authentication'
responses:
  '200':
    description: 'Ok. Show Log-in Page'

post:
  operationId: R102
  summary : 'R102: Login Action'
  description: 'Processes the login form submission. Access: PUB'
  tags:
    - 'M01: Authentication'
  requestBody:
    required: true
    content:
      application/x-www-form-urlencoded:
        schema:
          type: object
          properties:
            email:
              type: string
              description: 'The email of the user'
            password:
              type: string
              description: 'The password of the user'
          required:
            - email
            - password

  responses:
    '302':
      description: 'Redirect after login'
      headers:
        Location:
          description: 'URL for redirection'
          schema:
            type: string
          examples:
            302Success:
              description: 'Successful authentication. Redirect to user profile.'
              value: '/'
            302Error:
              description: 'Failed authentication. Redirect to Login Page.'
              value: '/login'

/logout:
  post:
    operationId: R103
    summary: 'R103: Logout Action'
    description: 'Processes the logout action. Access: USR'
    tags:
      - 'M01: Authentication'
    responses:
      '302':
        description: 'Redirect to Home Page'
        headers:
          Location:
            schema:
              type: string
            examples:
              302Success:
                description: 'Successful logout. Redirect to login page.'
                value: '/'
              302Error:
```

```
description: 'Failed logout. Redirect to Home Page'
value: '/'
```

/register:

get:

```
operationId: R104
summary: 'R104: Register Form'
description: 'Provide register form. Access: PUB'
tags:
  - 'M01: Authentication'
```

responses:

```
'200':
  description: 'Ok. Show Register Page'
```

post:

```
operationId: R105
summary: 'R105: Register Action'
description: 'Processes the register form submission. Access: PUB'
tags:
  - 'M01: Authentication'
```

requestBody:

required: true

content:

application/x-www-form-urlencoded:

schema:

type: object

properties:

name:

type: string

description: 'The name of the user'

username:

type: string

description: 'The username of the user'

email:

type: string

description: 'The email of the user'

password:

type: string

description: 'The password of the user'

password_confirmation:

type: string

description: 'The password confirmation of the user'

required:

- name
- email
- password
- password_confirmation

responses:

```
'302':
  description: 'Redirect after registration'
```

headers:

Location:

schema:

type: string

examples:

302Success:

description: 'Successful registration. Redirect to user profile.'

value: '/'

302Error:

description: 'Failed registration. Redirect to Register Page'

value: '/register'

MODULE 02: Individual Profile and User administration

/profile/{username}:

get:

```
operationId: R201
summary: 'R201: View Profile'
description: 'Provide user profile page. Access: PUB'
tags:
  - 'M02: Individual Profile and User administration'
parameters:
  - in: path
    name: username
    description: 'The username of the user'
    schema:
      type: string
    required: true
responses:
  '200':
    description: 'Ok. Show User Profile Page'
  '404':
    description: 'User not found'
```

```
delete:
  operationId: R202
  summary: 'R202: Delete User'
  description: 'Delete user. Access: ADM'
  tags:
    - 'M02: Individual Profile and User administration'
  parameters:
    - in: path
      name: username
      description: 'The username of the user'
      schema:
        type: string
      required: true
  responses:
    '202':
      description: 'Accepted. User deleted'
    '404':
      description: 'User not found'
```

```
/profile/edit:
  get:
    operationId: R203
    summary: 'R203: Edit Profile'
    description: 'Provide user profile edit form. Access: OWN, ADM'
    tags:
      - 'M02: Individual Profile and User administration'
    responses:
      '200':
        description: 'Ok. Show User Profile Edit Page'
  patch:
    operationId: R204
    summary: 'R204: Update Profile'
    description: 'Processes the user profile edit form submission. Access: OWN, ADM'
    tags:
      - 'M02: Individual Profile and User administration'
    requestBody:
      required: true
      content:
        application/x-www-form-urlencoded:
          schema:
            type: object
            properties:
              name:
                type: string
                description: 'The name of the user'
              email:
                type: string
                description: 'The email of the user'
              bio:
```

```

    type: string
    description: 'The bio of the user'
  example:
    name: 'John Doe'
    email: 'john@example.com'
    bio: 'I am a software engineer'
  responses:
    '302':
      description: 'Redirect after profile update'
      headers:
        Location:
          schema:
            type: string
      examples:
        302Success:
          description: 'Successful profile update. Redirect to user profile.'
          value: '/profile/{username}'
        302Error:
          description: 'Failed profile update. Redirect to Edit Profile Page'

/profile/search:
  get:
    operationId: R205
    summary: 'R205: Search User'
    description: 'Provide user search page. Access: ADM'
    tags:
      - 'M02: Individual Profile and User administration'
    parameters:
      - in: query
        name: username
        description: 'The username of the user'
        schema:
          type: string
        required: true
    responses:
      '200':
        description: 'Ok. Show User Search Page'

/profile/{username}/ban:
  patch:
    operationId: R206
    summary: 'R206: Ban User'
    description: 'Ban user. Access: ADM'
    tags:
      - 'M02: Individual Profile and User administration'
    parameters:
      - in: path
        name: username
        description: 'The username of the user'
        schema:
          type: string
        required: true
    responses:
      '202':
        description: 'Accepted. User banned'
      '404':
        description: 'User not found'

# MODULE 03: Publications

/questions/create:
  get:
    operationId: R301
    summary: 'R301: Create Question Form'
    description: 'Provide create question form. Access: USR'
    tags:
```

```
- 'M03: Publications'
responses:
  '200':
    description: 'Ok. Show Create Question Page'
/questions:
  post:
    operationId: R302
    summary: 'R302: Create Question Action'
    description: 'Processes the create question form submission. Access: USR'
    tags:
      - 'M03: Publications'
    requestBody:
      required: true
      content:
        application/x-www-form-urlencoded:
          schema:
            type: object
            properties:
              title:
                type: string
                description: 'The title of the question'
              content:
                type: string
                description: 'The body of the question'
            required:
              - title
              - content
            example:
              title: 'How to create a web API?'
              content: 'I am trying to create a web API but I am having some trouble. Can someone help me?'
    responses:
      '302':
        description: 'Redirect after question creation'
        headers:
          Location:
            schema:
              type: string
            examples:
              302Success:
                description: 'Successful question creation. Redirect to question page.'
                value: '/'
              302Error:
                description: 'Failed question creation. Redirect to Create Question Page'
                value: '/questions/create'
/questions/search:
  get:
    operationId: R303
    summary: 'R303: Search Questions'
    description: 'Provide search questions page. Access: PUB'
    tags:
      - 'M03: Publications'
    parameters:
      - in: query
        name: query
        description: 'Search query'
        schema:
          type: string
        required: true
    responses:
      '200':
        description: 'Ok. Show Search Questions Page'
/questions/{id}:
  get:
    operationId: R304
    summary: 'R304: View Question'
    description: 'Provide question page. Access: PUB'
    tags:
```



```
- 'M03: Publications'
parameters:
- in: path
  name: id
  description: 'The id of the question'
  schema:
    type: integer
  required: true
responses:
  '200':
    description: 'Ok. Show Question Page'
  '404':
    description: 'Question not found'
delete:
  operationId: R305
  summary: 'R305: Delete Question'
  description: 'Delete question. Access: OWN'
  tags:
    - 'M03: Publications'
  parameters:
    - in: path
      name: id
      description: 'The id of the question'
      schema:
        type: integer
      required: true
  responses:
    '302':
      description: 'Redirect after question deletion'
      headers:
        Location:
          schema:
            type: string
      examples:
        302Success:
          description: 'Successful question deletion. Redirect to Home Page.'
          value: '/'
        302Error:
          description: 'Failed question deletion. Redirect to Question Page'
          value: '/questions/{id}'
put:
  operationId: R306
  summary: 'R306: Update Question'
  description: 'Processes the question edit form submission. Access: OWN'
  tags:
    - 'M03: Publications'
  parameters:
    - in: path
      name: id
      description: 'The id of the question'
      schema:
        type: integer
      required: true
  requestBody:
    required: true
    content:
      application/x-www-form-urlencoded:
        schema:
          type: object
          properties:
            title:
              type: string
              description: 'The title of the question'
            content:
              type: string
              description: 'The body of the question'
          required:
```

```
      - title
      - content
    example:
      title: 'How to create a web API?'
      content: 'I am trying to create a web API but I am having a trouble with sql. Can someone help me?'
responses:
  '302':
    description: 'Redirect after question update'
    headers:
      Location:
        schema:
          type: string
        examples:
          302Success:
            description: 'Successful question update. Redirect to question page.'
            value: '/questions/{id}'
          302Error:
            description: 'Failed question update. Redirect to Edit Question Page'
            value: '/questions/{id}/edit'

/questions/{id}/edit:
  get:
    operationId: R307
    summary: 'R307: Edit Question'
    description: 'Provide question edit form. Access: OWN'
    tags:
      - 'M03: Publications'
    parameters:
      - in: path
        name: id
        description: 'The id of the question'
        schema:
          type: integer
        required: true
    responses:
      '200':
        description: 'Ok. Show Edit Question Page'

/questions/{id}/answer/create:
  get:
    operationId: R308
    summary: 'R308: Create Answer Form'
    description: 'Provide create answer form. Access: USR'
    tags:
      - 'M03: Publications'
    parameters:
      - in: path
        name: id
        description: 'The id of the question'
        schema:
          type: integer
        required: true
    responses:
      '200':
        description: 'Ok. Show Create Answer Page'

/answers:
  post:
    operationId: R309
    summary: 'R309: Create Answer Action'
    description: 'Processes the create answer form submission. Access: USR'
    tags:
      - 'M03: Publications'
    requestBody:
      required: true
      content:
        application/x-www-form-urlencoded:
          schema:
```



```

    type: object
    properties:
      content:
        type: string
        description: 'The body of the answer'
    required:
      - content
    example:
      content: 'You can create a web API by following these steps...'
responses:
  '302':
    description: 'Redirect after answer creation'
    headers:
      Location:
        schema:
          type: string
    examples:
      302Success:
        description: 'Successful answer creation. Redirect to question page.'
        value: '/questions/{id}'
      302Error:
        description: 'Failed answer creation. Redirect to Create Answer Page'
        value: '/questions/{id}/answer/create'

/answers/{id}:
  put:
    operationId: R310
    summary: 'R310: Update Answer'
    description: 'Processes the answer edit form submission. Access: OWN'
    tags:
      - 'M03: Publications'
    parameters:
      - in: path
        name: id
        description: 'The id of the answer'
        schema:
          type: integer
        required: true
    requestBody:
      required: true
      content:
        application/x-www-form-urlencoded:
          schema:
            type: object
            properties:
              content:
                type: string
                description: 'The body of the answer'
            required:
              - content
          example:
            content: 'You can create a web API by following these steps...'
    responses:
      '302':
        description: 'Redirect after answer update'
        headers:
          Location:
            schema:
              type: string
        examples:
          302Success:
            description: 'Successful answer update. Redirect to question page.'
            value: '/questions/{id}'
          302Error:
            description: 'Failed answer update. Redirect to Edit Answer Page'
            value: '/answers/{id}/edit'

  delete:
```

```
operationId: R311
summary: 'R311: Delete Answer'
description: 'Delete answer. Access: OWN'
tags:
  - 'M03: Publications'
parameters:
  - in: path
    name: id
    description: 'The id of the answer'
    schema:
      type: integer
    required: true
responses:
  '302':
    description: 'Redirect after answer deletion'
    headers:
      Location:
        schema:
          type: string
    examples:
      302Success:
        description: 'Successful answer deletion. Redirect to question page.'
        value: '/questions/{id}'
      302Error:
        description: 'Failed answer deletion. Redirect to question page'
        value: '/questions/{id}'

/answers/{id}/edit:
  get:
    operationId: R312
    summary: 'R312: Edit Answer'
    description: 'Provide answer edit form. Access: OWN'
    tags:
      - 'M03: Publications'
    parameters:
      - in: path
        name: id
        description: 'The id of the answer'
        schema:
          type: integer
        required: true
    responses:
      '200':
        description: 'Ok. Show Edit Answer Page'

# MODULE 04: Static pages
/:
  get:
    operationId: R401
    summary: 'R401: Home Page'
    description: 'Provide home page. Access: PUB'
    tags:
      - 'M04: Static pages'
    responses:
      '202':
        description: 'Ok. Show Home Page'

/aboutus:
  get:
    operationId: R402
    summary: 'R402: About Us Page'
    description: 'Provide about us page. Access: PUB'
    tags:
      - 'M04: Static pages'
    responses:
      '202':
        description: 'Ok. Show About Us Page'
```

```
/mainfeatures:
  get:
    operationId: R403
    summary: 'R403: Main Features Page'
    description: 'Provide main features page. Access: PUB'
    tags:
      - 'M04: Static pages'
    responses:
      '202':
        description: 'Ok. Show Main Features Page'

/contactus:
  get:
    operationId: R404
    summary: 'R404: Contact Us Page'
    description: 'Provide contact us page. Access: PUB'
    tags:
      - 'M04: Static pages'
    responses:
      '202':
        description: 'Ok. Show Contact Us Page'
```

A8: Vertical prototype

The artifact show the implemented features and the instructions to run our prototype.

1. Implemented Features

1.1. Implemented User Stories

User Story reference	Name	Priority	Responsible	Description
US01	View Top Questions	High	Rafael	As an user, I want to view the top questions so that I can see relevant topics
US02	Browse Questions	High	Rafael	As an user, I want to browse questions, so that I can find questions that interest me
US04	Full-text Search	High	Diogo	As an user, I want to do a full-text Search, so that I can find a specific content
US17	Register	High	Daniel	As a visitor, i want to register, so that i can be a member
US18	Log in	High	Daniel	As a visitor, i want to log in, so that i can access to special features, only available to members
US19	Logout	High	Rafael	As a member,I want to logout ,so that my account stays safe
US20	View Profile	High	Tiago	As a member,I want to view profile ,so that i can keep my details up to date
US21	Edit Profile	High	Tiago	As a member,I want to edit profile ,so that i can keep my details up to date
US22	View Personal Feed	High	Rafael	As a member,I want to view personal feed ,so that i can see content based on my preferences
US23	Post Question	High	Tiago	As a member,I want to post a question ,so that i can have answers to solve my doubts/problems
US24	Post Answer	High	Tiago	As a member,I want to post a answer,so that i can help other users
US25	View My Questions	High	Daniel	As a member,I want to view my questions,so that i can track answers
US26	View My Answers	High	Daniel	As a member,I want to view my answers ,so that i can see if it helped other users

User Story reference	Name	Priority	Responsible	Description
US39	Edit Question	High	Diogo	As an author,i want to edit a question ,so that i can add some details to my post
US40	Delete Question	High	Diogo	As an author,i want to delete a question,so that i can remove content that is no longer important
US41	Edit Answer	High	Diogo	As an author,i want to edit a answer ,so that i can improve it or change some aspects
US42	Delete Answer	High	Diogo	As an author,i want to delete a answer ,so that i can remove it if it isn't useful
US47	Administer User Accounts	High	Tiago	As an admin,I want to search, view, edit, and create user accounts,so that i can manage each user account

1.2. Implemented Web Resources

Module M01: Authentication

Web Resource Reference	URL
R101: Login Form	http://localhost:8001/login
R102: Login Action	POST Login
R103: Logout Action	POST Logout
R104: Register Form	http://localhost:8001/register
R105: Register Action	POST Register

Module M02: Individual Profile and User Administration

Web Resource Reference	URL
R201: View Profile	http://localhost:8001/profile/{username} .
R202: Delete User	DELETE User
R203: Edit Profile	http://localhost:8001/profile/edit
R204: Update Profile	PATCH Edit Profile
R205: Search User	http://localhost:8001/profile/search?query={userinput} .
R206: Ban User	PATCH Ban

Module M03: Publications

Web Resource Reference	URL
R301: Create Question Form	http://localhost:8001/questions/create
R302: Create Question Action	POST Question
R303: Search Questions	http://localhost:8001/questions/search?query={userinput} .
R304: View Question	http://localhost:8001/questions/{id} .
R305: Delete Question	DELETE Question
R306: Update Question	PUT Question
R307: Edit Question	http://localhost:8001/questions/{id}/edit
R308: Create Answer Form	GET Answer form
R309: Create Answer Action	POST Answer

Web Resource Reference	URL
R310: Update Answer	PUT Answer
R311: Delete Answer	DELETE Answer
R312: Edit Answer	http://localhost:8001/answers/{id}/edit

Module M04: Static Pages

Web Resource Reference	URL
R401: View home page	http://localhost:8001/home

2. Prototype

Command to start Docker image:

```
docker run -d --name lbaw24141 -p 8001:80 gitlab.up.pt:5050/lbaw/lbaw2425/lbaw24141
```

Then the website prototype is available at [Prototype Implementation](#)

Credentials to log in:

- **Normal User:**
 - Email: user_with_questions@example.com
 - Password: 1234
- **Banned User:**
 - Email: banned_user@example.com
 - Password: 1234
- **Admin:**
 - Email: admin@example.com
 - Password: 1234

All source code is available at the [Group Repository](#).

Revision history

Changes made to the first submission:

1. Item 1
2. ..

GROUP24141, 09/10/2024

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